Task 1 Natural Language Processing Skill

1. A company is building a Resume Screening System. The system receives raw resume text and must preprocess it to extract meaningful keywords for ranking candidates. You are required to build an NLP pipeline that performs the following steps:

Tasks: 1. Tokenize the input text into words 2. Remove stopwords 3. Perform stemming 4. Perform lemmatization 5. Apply POS tagging 6. Compute frequency distribution of remaining words

```
import nltk

nltk.download('punkt_tab')

nltk.download('stopwords')

nltk.download('wordnet')

nltk.download('averaged_perceptron_tagger')


resume_text = "John is a Python Developer with 3 years of experience in Machine Learning,Data Analysis, Natural Language Processing, and Web Development.Skilled in Python, SQL, Pandas, NumPy, Scikit-Learn, and TensorFlow."


# 1. Tokenization

tokens = word_tokenize(resume_text)

print("Tokens:")

print(tokens)
```

```
# 2. Stopword Removal
stop_words = set(stopwords.words('english'))


filtered_words = [
    word.lower() for word in tokens
    if word.isalpha() and word.lower() not in stop_words
]


print("\nAfter Stopword Removal:")
print(filtered_words)


# 3. Stemming
stemmer = PorterStemmer()


stemmed_words = [stemmer.stem(word) for word in filtered_words]


print("\nStemmed Words:")
print(stemmed_words)


# 4. Lemmatization
lemmatizer = WordNetLemmatizer()
```

```
lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]


print("\nLemmatized Words:")

print(lemmatized_words)


# 5. POS Tagging

pos_tags = nltk.pos_tag(lemmatized_words)


print("\nPOS Tags:")

print(pos_tags)


# 6. Frequency Distribution

freq_dist = FreqDist(lemmatized_words)


print("\nFrequency Distribution:")

for word, freq in freq_dist.items():

    print(word, ":", freq)
```

```
Tokens:
['John', 'is', 'a', 'Python', 'Developer', 'with', '3', 'years', 'of', 'experience', 'in', 'Machine', 'Learning', ',', 'Data', 'Analysis', ',', 'Natura
l', 'Language', 'Processing', ',', 'and', 'Web', 'Development.Skilled', 'in', 'Python', ',', 'SQL', ',', 'Pandas', ',', 'NumPy', ',', 'Scikit-Learn',
',', 'and', 'TensorFlow', '.']

After Stopword Removal:
['john', 'python', 'developer', 'years', 'experience', 'machine', 'learning', 'data', 'analysis', 'natural', 'language', 'processing', 'web', 'python',
'sql', 'pandas', 'numpy', 'tensorflow']

Stemmed Words:
['john', 'python', 'develop', 'year', 'experi', 'machin', 'learn', 'data', 'analysi', 'natur', 'languag', 'process', 'web', 'python', 'sql', 'panda',
'numpi', 'tensorflow']

Lemmatized Words:
['john', 'python', 'developer', 'year', 'experience', 'machine', 'learning', 'data', 'analysis', 'natural', 'language', 'processing', 'web', 'python',
'sql', 'panda', 'numpy', 'tensorflow']

POS Tags:
[('john', 'NN'), ('python', 'NN'), ('developer', 'IN'), ('year', 'NN'), ('experience', 'NN'), ('machine', 'NN'), ('learning', 'VBG'), ('data', 'NNS'),
('analysis', 'NN'), ('natural', 'JJ'), ('language', 'NN'), ('processing', 'NN'), ('web', 'NN'), ('python', 'NN'), ('sql', 'NN'), ('panda', 'NN'), ('num
py', 'RB'), ('tensorflow', 'VB')]

Frequency Distribution:
john : 1
python : 2
developer : 1
year : 1
experience : 1
machine : 1
learning : 1
data : 1
analysis : 1
natural : 1
language : 1
processing : 1
web : 1
sql : 1
```